

A Piecewise Approach to Mapping Interactions between Room-scale Environments in Remote Mixed/Augmented Reality

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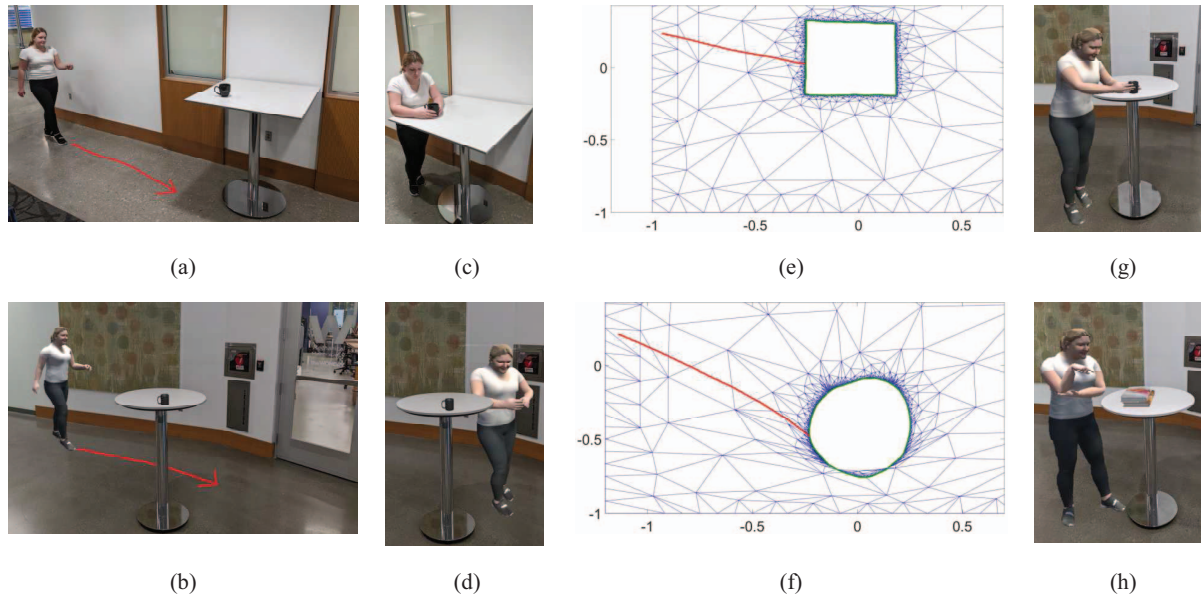


Figure 1: The motivation for the necessity of mapping interactions between remote environments (a)-(d): (a) and (c) represent the user in their environment walking towards a table to interact with an object. (b) and (d) represent the user via their photorealistic avatar in the remote environment with a direct mapping of the interaction, leading to an error and necessitating the redirection in mapping. (e)-(h) represents our proposed piecewise approach to obtain the mapping of interaction for room scale environments. We map the locomotion in 2D, using mesh deformation, i.e., deform the mesh associated with the user's environment (e), to the spatial constraints of the avatar's environment. Once the avatar reaches the correct equivalent position in its environment, its kinematic structure is redirected in 3D depending on the interaction: positional contact interaction (g) or pointing interaction (h). Note that the interaction is redirected by preserving the pose of the original interaction.

ABSTRACT

Naturalistic interactions from remote settings require the user to have the ability to move and interact with their environment. However, a direct mapping of the interactions onto their photorealistic avatars will cause errors, primarily due to the different spatial configurations between the environments. Hence, the interactions need to be redirected, naturalistically. This redirection is a computationally intensive process and current techniques can only be used for localized spaces. We propose a piecewise approach, where the interaction mapping is split into a 2D locomotion and a 3D gesture redirection mechanism, reducing the

computational requirement and making it applicable to room-scale environments.

Index Terms: Human-centered computing-Human computer interaction (HCI)—Interaction paradigms--Mixed/augmented reality

1 INTRODUCTION

Mixed/Augmented reality (MR/AR) has the potential to simulate in-person interactions between users and their environments from remote settings. Naturalistic interactions from such settings involve having the ability to move around and interact with the environment, similar to a physically present interaction. However, the interactions cannot be directly mapped between the user's and avatar's environments as it leads to an interaction error. Such errors cause a loss of meaning in the interaction (Fig. 1 (a)-(d)) and is primarily caused due to the difference in spatial configurations between the remote and local environments. Thus, interactions need to be redirected to correct for the error, but also in a way that it preserves the nonverbal body

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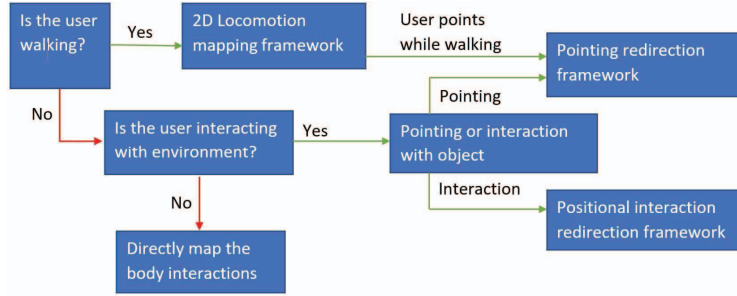


Figure 2: The piecewise framework for mapping user's interaction to their avatar environments.

pose of the original interaction. This involves applying real time optimization techniques that are computationally intensive [1]. Current works [1] [2] can accomplish this only for localized spaces. These methods cannot be directly applied to large room-scale environments due to high computational cost. In this work, we propose a piecewise approach to this problem, i.e., we divide the mapping redirection into a 2D locomotion component and a 3D gesture redirection component. In this way majority of the room scale redirection takes place in 2D which is computationally less intensive. The details of the framework are provided below.

2 PIECEWISE MAPPING FRAMEWORK

In the framework (Fig. 2), whenever the user is walking, the equivalent path for the avatar to follow is mapped in 2D in the remote environment. The avatar's velocity profile and walking animation gait is varied accordingly for naturalistic locomotion. If the user points at an object, the avatar's pointing will be redirected according to the configuration of its environment (Fig. 1(h)). Similarly, whenever the user has contact interactions with an object, the avatar's actions are redirected in 3D (Fig. 1(g)) [1]. If there is no interaction with the environment, the user's interactions are directly mapped onto the avatar. Thus, most of the room scale interactions are in 2D and 3D interactions happen only to redirect gestures which reduces the overall computational cost. The main two mapping mechanisms of the framework are described in more detail below:

2.1 2D Locomotion mapping

A room environment R can be considered to consist of an external boundary E , interior furniture objects $\{I_1, I_2 \dots I_n\}$ and, walkable spaces W in the interior (Fig.1 (e) and (f)). When a user walks, their path can be tracked as a set of Cartesian position coordinates (x, y, z) . These coordinates can be obtained from the head mounted display (HMD) the user wears. For the avatar to reach the same corresponding location as the user in their environment, the path in environment R_1 (Fig. 1(e)) has to be modified according to the constraints of environment R_2 (Fig. 1(f)). Thus, the problem becomes one of finding a mapping function between the room environments 1 and 2, $f_R: R_1 \rightarrow R_2$ for all possible paths. This will include finding the corresponding equivalent maps for the walkable surfaces $f_w: W_1 \rightarrow W_2$, exterior room boundaries $f_E: E_1 \rightarrow E_2$, and for each furniture object between the environments $f_I: \{I_{1_1}, I_{1_2} \dots I_{1_n}\} \rightarrow \{I_{2_1}, I_{2_2} \dots I_{2_n}\}$. To obtain f_I we use non-rigid transformation techniques[3] for each furniture object and to obtain f_w we use the mesh deformation that minimizes the Dirichlet energy associated

with the W_1 mesh, subject to the boundary conditions of the W_2 mesh.

2.2 Gesture mapping

When the user interacts with the environment, it needs to be redirected according to the avatars environment while also preserving the pose of the original interaction. To satisfy this dual objective, we use a Multi-objective optimization (MOO) framework [1], where the primary objective will be to correct the positional/pointing error of the interaction and the secondary objective will be to preserve the body pose captured during the user's original interaction. If the user has no interaction with the environment, their interactions are directly mapped onto their avatar. An example of such an interaction would be when a user is stationary and talking to another user using hand gestures.

3 CONCLUSION AND FUTURE WORK

In this work, a piecewise approach to mapping interactions for room scale remote environments in AR is proposed. This approach requires lesser computation compared to current approaches. For future work we will implement the proposed framework whose components include developing a real time method to obtain 2D meshes of room environments, development of the trajectory planner for modifying the avatar's velocity profile and integrating the locomotion and gesture redirection frameworks. We will also conduct a pilot study to test the framework when users perform a task in AR.

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